RGIPT

Subject: DevOps

Class: B.Tech 2nd year

Time Allowed: 45 minutes **Maximum Marks: 6**

All questions are compulsory unless stated otherwise.

Name: ___________________________

Roll Number: ____________________

Class: B.Tech 2nd year Section: ________

Section A: Short Questions

Answer all questions. Each question carries 2 marks.

1. [Easy] Explain the difference between `docker run` with the `-d` flag and without it, and describe a scenario where each is appropriate. [2 Marks]

2. [Moderate] Describe the steps to remove all stopped containers, unused images, and unused volumes in a single cleanup operation. Include the specific Docker commands. [2 Marks]

3. [Challenging] A developer needs to run a web application container that listens on port 8080 inside the container and should be accessible on host port 80. Additionally, the container must have an environment variable `ENV=production` and a bind mount from host `/var/www/html` to container `/usr/share/nginx/html`. Write the complete `docker run` command achieving this, and explain each option used. [2 Marks]

End of Question Paper

Answer Key:

1. `docker run` without `-d` runs the container in the foreground, attaching the terminal to the container's STDIN/STDOUT, which is useful for debugging or when interactive input is needed (e.g., running a shell). `docker run -d` runs the container in detached mode (background) and returns the container ID immediately; it is suitable for services that should keep running independently, such as a web server.

2. 1. Remove stopped containers: `docker container prune -f` (or `docker rm $(docker ps -a -q)`). 2. Remove unused images: `docker image prune -a -f`. 3. Remove unused volumes: `docker volume prune -f`. These three prune commands together clean up all stopped containers, dangling/unused images, and orphaned volumes.

3. `docker run -d -p 80:8080 -e ENV=production -v /var/www/html:/usr/share/nginx/html --name webapp ` Explanation: - `-d` runs the container in detached (background) mode. - `-p 80:8080` maps host port 80 to container port 8080, exposing the service. - `-e ENV=production` sets the environment variable

`ENV` inside the container. - `-v /var/www/html:/usr/share/nginx/html` creates a bind mount, linking the host directory to the container path. - `--name webapp` gives the container a readable name. - `` is the Docker image to instantiate.